SLE-4: ARCHITECTURE DECISION RECORD (ADR)

02AML204 – Introduction to Artificial Intelligence

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PART A: ARCHITECTURE DECISION RECORD

ADR Title: Using BFS-1 and DFS-1 for the Same 32-Node Graph Search Problem

1. Context

The project is a Graph Search System designed to find a path from a start node S to a goal node AF in a 32-node graph. Two search algorithms, BFS-1 and DFS-1, are implemented and evaluated on the same graph. The system tracks visited nodes to avoid processing the same node repeatedly and checks whether the current node is the goal.

2. Decision

The system will implement and evaluate both BFS-1 (Breadth-First Search) and DFS-1 (Depth-First Search) using the same graph, start node S, and goal node AF. The common search responsibilities include visited-node tracking, goal checking, and producing the search result. This keeps the comparison consistent and supports the architecture defined for the project.

3. Alternatives Considered

- Use only BFS-1 for the graph search.
- Use only DFS-1 for the graph search.
- Use another search algorithm instead of comparing BFS-1 and DFS-1.

4. Rationale / Justification

Using both BFS-1 and DFS-1 allows the project to demonstrate two standard graph-search strategies under the same test conditions. The SLE-2 observations showed that BFS-1 explored 32 nodes while DFS-1 explored 11 nodes for the tested graph. The recorded average execution times were 0.005897 ms for BFS-1 and 0.002954 ms for DFS-1. These observations support comparing both approaches, while the results are understood to be specific to the tested graph and node ordering rather than a universal claim that one algorithm is always faster.

5. Consequences

Positive consequences:

- The system supports two different graph-search strategies.
- Both algorithms can be compared under the same graph and goal conditions.
- The architecture separates BFS-1 and DFS-1 responsibilities clearly.
- The result can be evaluated using observed search and execution measurements.

Trade-offs:

- Implementing two algorithms requires more code and testing than using one.
- BFS-1 and DFS-1 can behave differently depending on graph structure and node ordering.
- Measured execution time on a small graph should not be generalized to all graphs.

PART B: SHORT SUMMARY OF MY SLE JOURNEY

SLE-1: Started the AI project and understood the problem statement and basic requirements for the Graph Search System.

SLE-2: Implemented and evaluated BFS-1 and DFS-1 on the 32-node graph. The recorded observations included explored nodes, execution time, profiling samples, and error status.

SLE-3: Designed the system architecture using the C4 approach and documented the main system structure, search components, visited-node tracking, goal checking, and result output.

SLE-4: Prepared the Architecture Decision Record to document the major design decision of using and comparing BFS-1 and DFS-1 on the same graph, together with its rationale and consequences.

AI Contribution

AI tools were used as an assistance resource during the SLE work for understanding requirements, organizing the ADR/report structure, improving wording, preparing documentation, and reviewing the presentation of the project. The project decisions, graph-search implementation, testing, measured results, and final responsibility remained with the student.

CONCLUSION

The ADR records the decision to implement and compare BFS-1 and DFS-1 for the same 32-node graph-search problem. The decision is supported by consistent testing conditions and the observations obtained during SLE-2, while the C4 architecture developed in SLE-3 provides a clear structure for the search system.